User-centered Design

Home of the Mozilla Project
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Wikipedia
http://www.mozilla.org/
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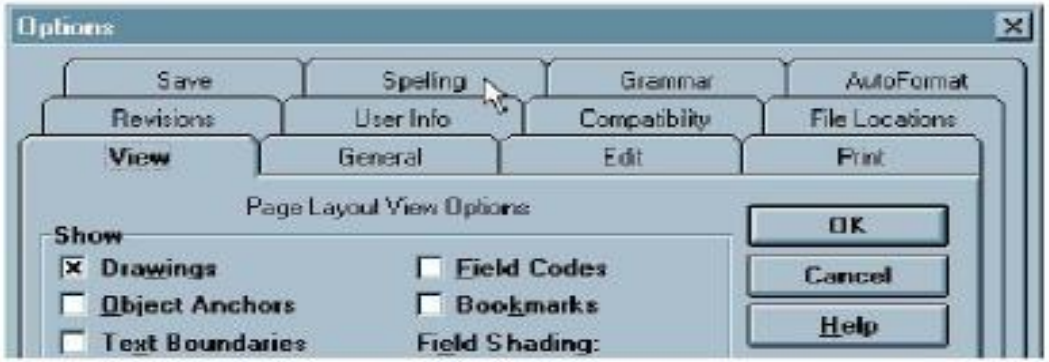

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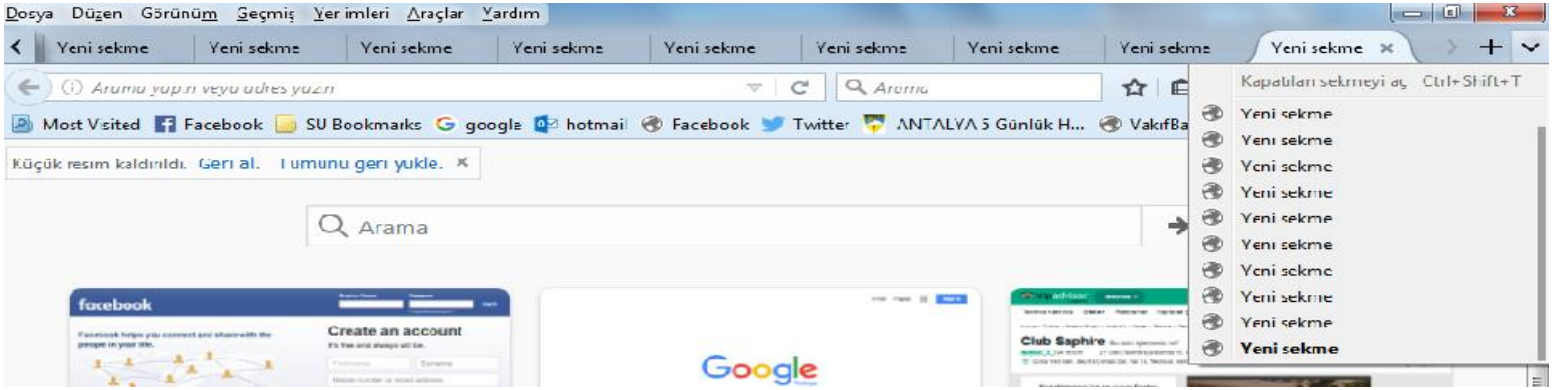

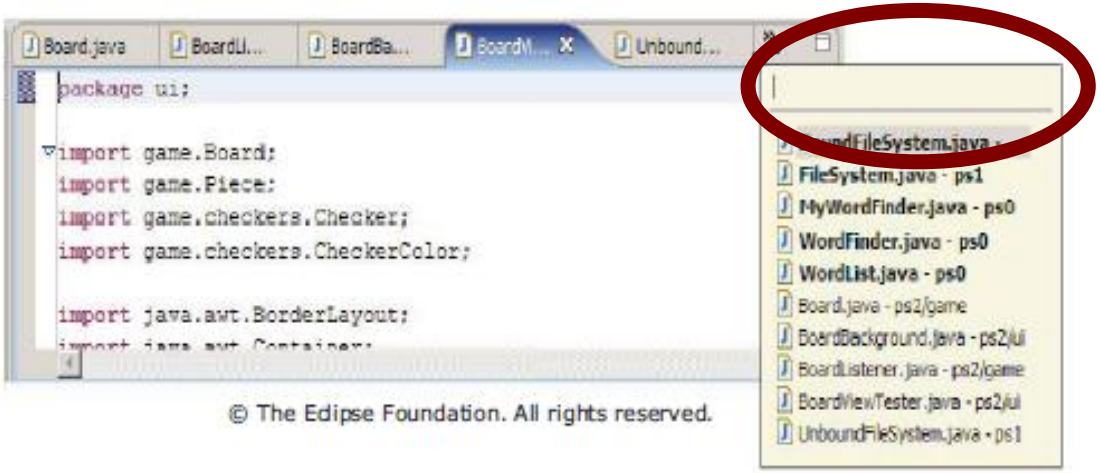

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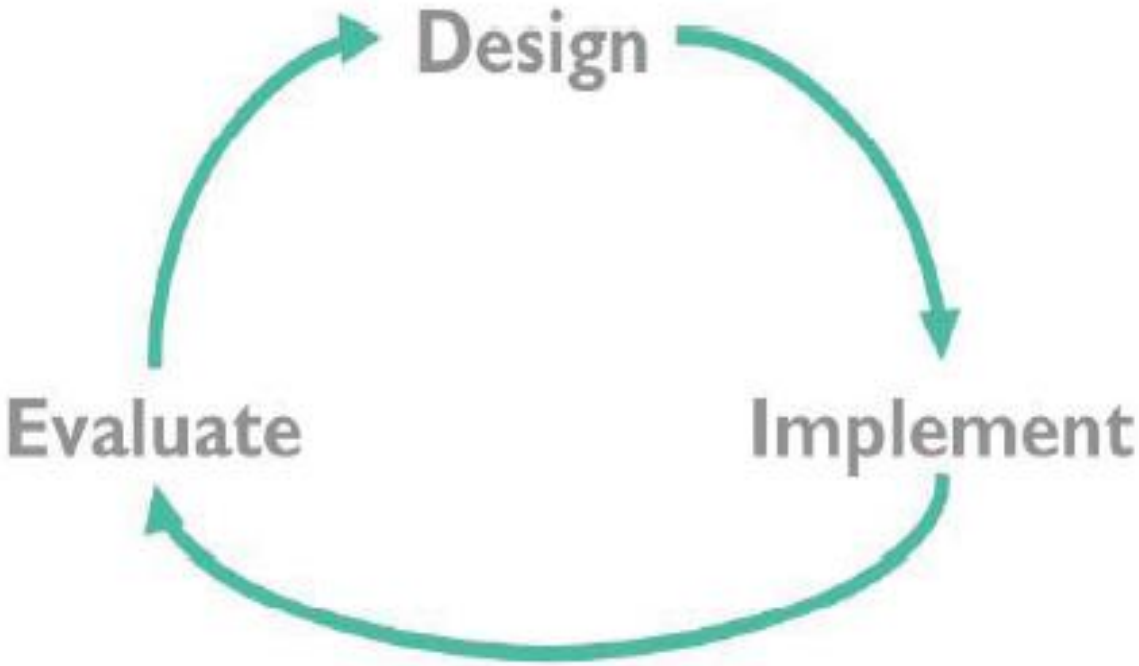
disregards the **natural, spatial mapping** that tabs provide

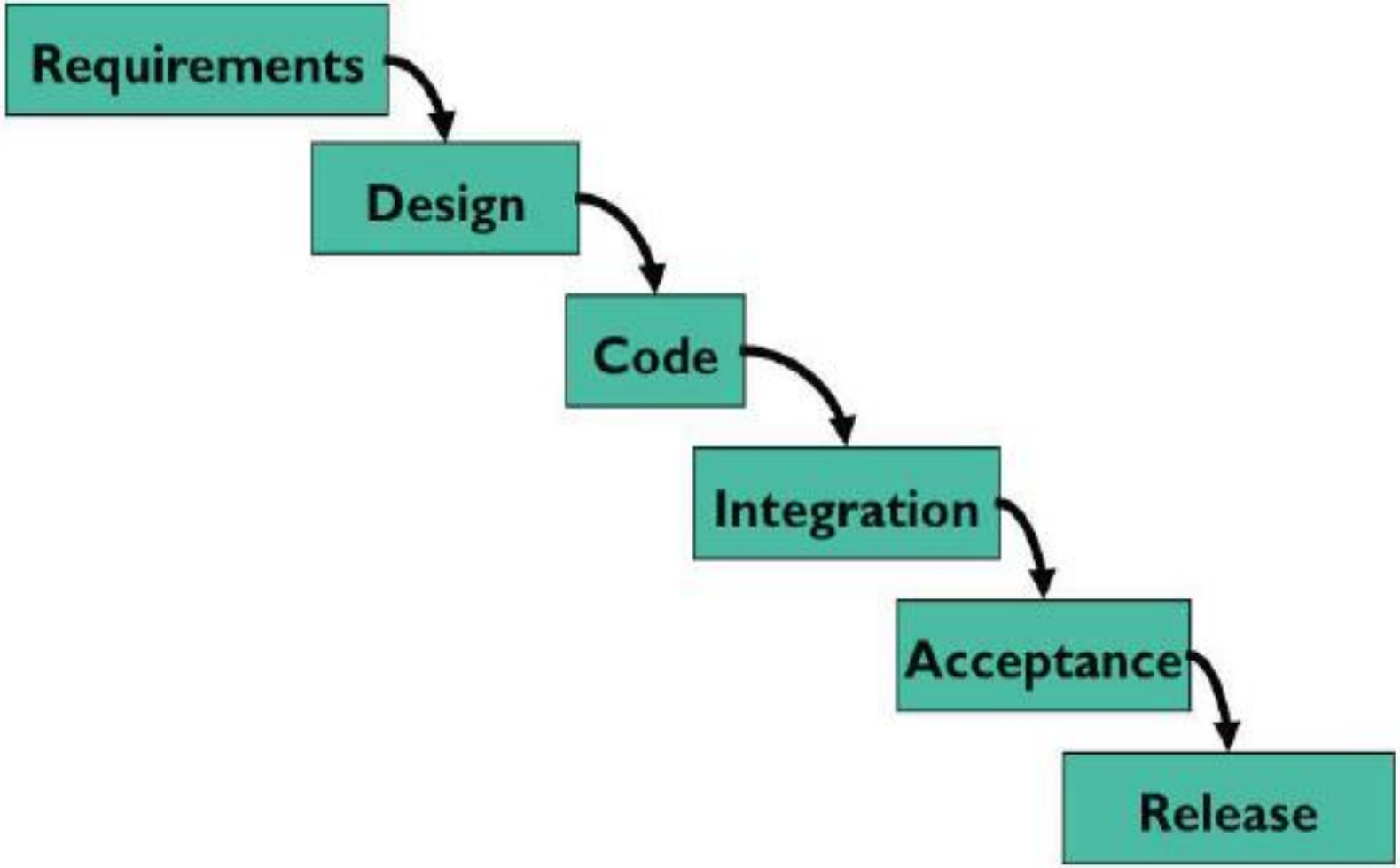
Today's Topics

- Iterative design
- User-centered design

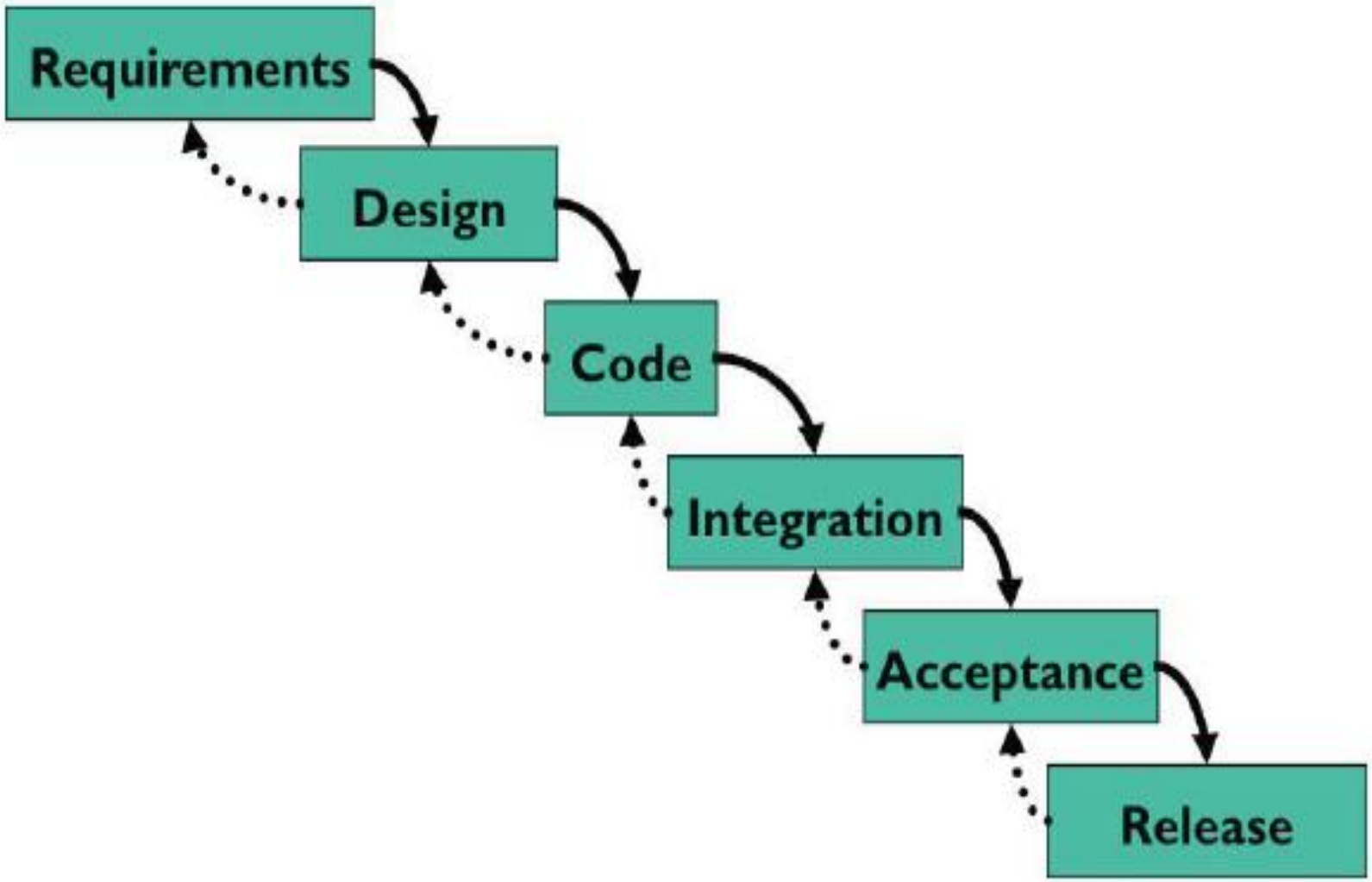
Iterative Design



Traditional Software Engineering Process: Waterfall Model



Feedback in the Waterfall Model



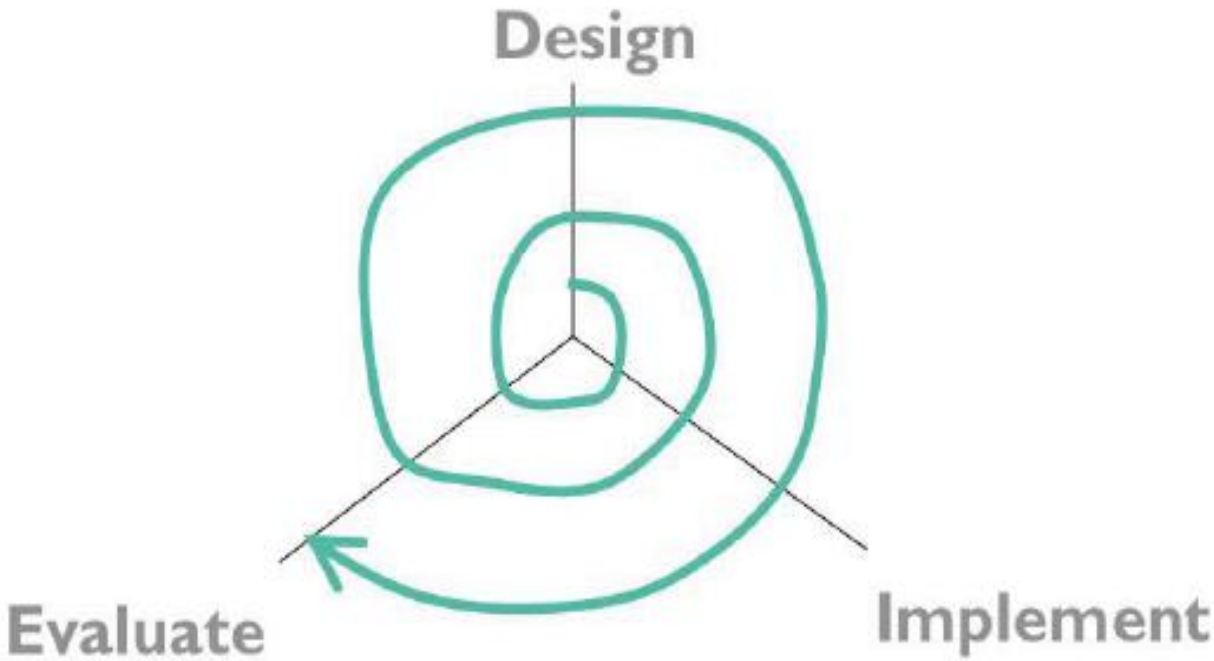
Waterfall Model Is Bad for UI Design

- User interface design is risky
  - So we're likely to get it wrong
- Users are not involved in validation until acceptance testing
  - So we won't find out until the end
- UI flaws often cause changes in requirements and design
  - So we have to throw away carefully-written and tested code

Iterative Design the Wrong Way

- Every iteration corresponds to a release
  - Evaluation (complaints) feeds back into next version's design
- Using your paying customers to evaluate your usability
  - They won't like it
  - They won't buy version 2

Spiral Model



radial dimension of the spiral model corresponds to the **cost** of the iteration step

Iterative Design of User Interfaces

- Early iterations use cheap prototypes
  - **Parallel design** is feasible: build & test multiple prototypes to explore design alternatives
- Later iterations use richer implementations, after UI risk has been mitigated
- More iterations generally means better UI
- Only mature iterations are seen by the world

User-Centered Design

- Iterative design
- Early focus on users and tasks
  - user analysis: who the users are
  - task analysis: what they need to do
  - involving users as evaluators, consultants, and sometimes designers
- Constant evaluation
  - Users are involved in every iteration
  - Every prototype is evaluated somehow

Design

- Task & user analysis
  - "Know thy user"
- Design principles
  - Learnability
  - Visibility
  - Efficiency
  - Error prevention and error handling
  - User control and freedom

Implementation

- Prototyping
  - Cheap, throw-away implementations
  - Low-fidelity: paper, Wizard of Oz
  - Medium-fidelity: HTML, Flash
- GUI implementation techniques
  - Output & input models
  - Desktop vs. web GUI
  - Constraints & layout

Evaluation

- Evaluation puts prototypes to the test
- Expert evaluation
  - Heuristics and walkthroughs
- Predictive evaluation
  - Testing against an engineering model (a simulated user)
- Empirical evaluation
  - Watching users do it

Summary

- Models for software development
  - Waterfall model makes sense for low-risk projects
  - Iterative or spiral models are needed when the requirements and design space are unknown or risky
  - UI development is often risky
- User-centered design process
  - Iterative, prototype-driven
  - Early focus on users and tasks
  - Constant evaluation